

Stress/Strain Relationships

Poisson's Ratio

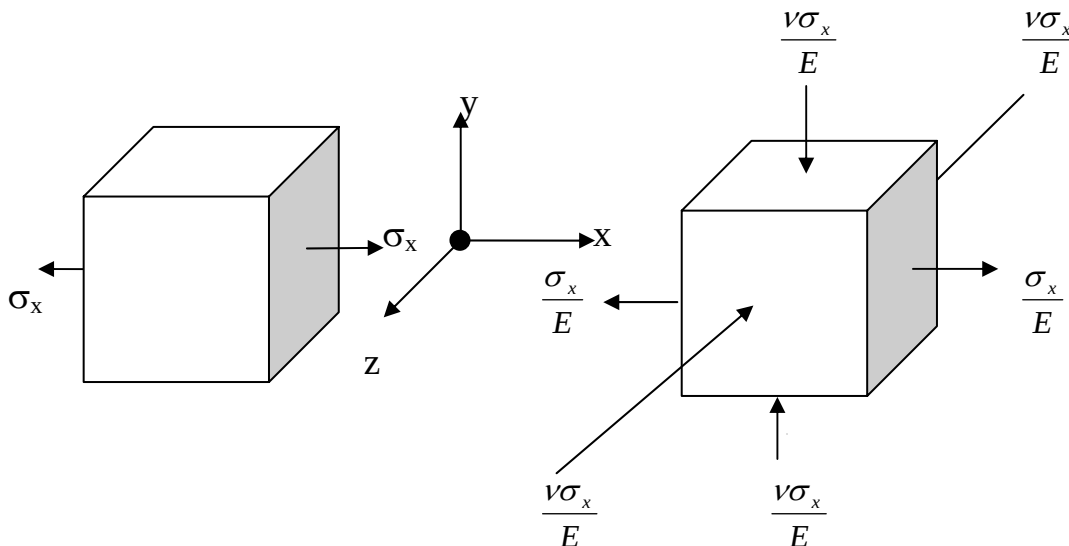


- The bar shown is being subjected to a longitudinal tensile stress.
- The bar will extend in the direction of stress and contract in the transverse or lateral directions. THE OPPOSITE IS TRUE FOR COMPRESSION!
- Lateral strain is proportional to the longitudinal strain, such that:

$$\text{Lateral strain} = -\nu \times \text{direct strain} \quad \text{or} \quad \nu = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$$

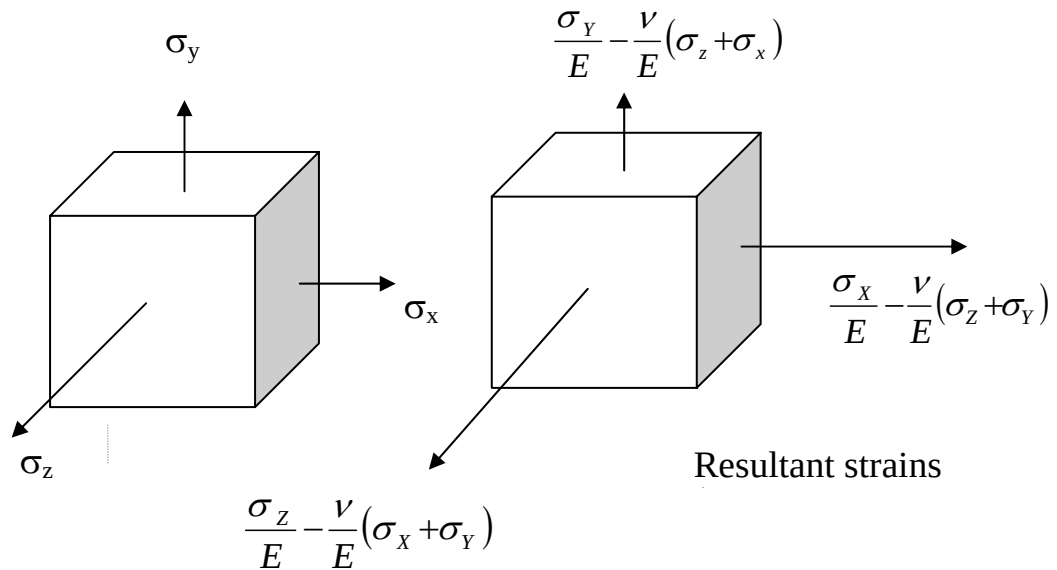
General Stress/Strain Relationships

- Consider an element of material subjected to a uniaxial stress, σ_x .



Resultant Strains

- Consider an element subjected to a tri-axial stress system σ_x , σ_y and σ_z .



Therefore, elastic stress/strain relationships can be written as:

Hooke's Laws

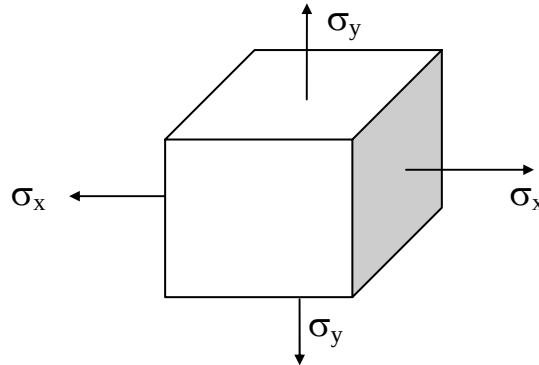
$$\varepsilon_{xx} = \frac{1}{E} \left(\sigma_{xx} - \nu (\sigma_{yy} + \sigma_{zz}) \right)$$

$$\varepsilon_{yy} = \frac{1}{E} \left(\sigma_{yy} - \nu (\sigma_{zz} + \sigma_{xx}) \right)$$

$$\varepsilon_{zz} = \frac{1}{E} \left(\sigma_{zz} - \nu (\sigma_{xx} + \sigma_{yy}) \right)$$

Plane Stress and Plane Strain

In many practical situations σ_z is zero, and this is referred to as a **PLANE STRESS** condition. Note however that ϵ_{zz} is **NOT** zero.



If strain in the z direction is zero this is referred to as **PLANE STRAIN**, however this does not imply zero stress in that direction.

Cylindrical and Spherical Pressure Vessels

Thin pressure vessels are a good example of plane stress and can be found in many applications, from pressurised containers through to pipelines. An aircraft fuselage can be considered to be a pressurised container much the same as a spray can or submarine pressure hull.

General Stress System in a Shell

In a 3-dimensional shell of arbitrary wall thickness, stresses are perpendicular to the thickness and in two principal orthogonal directions (tangential to the surface geometry). In **THICK** cylinders, these stresses have a non-uniform distribution through the thickness (or assumed constant). In **THIN** cylinders, each of these stresses can be assumed to be constant through the thickness (or negligible).

How do you decide whether a given pressure vessel is 'thin' or 'thick'?

If the wall thickness is less than $1/10$ of the radius of curvature of the shell, we can assume the shell is **thin**.

Thin walled assumptions

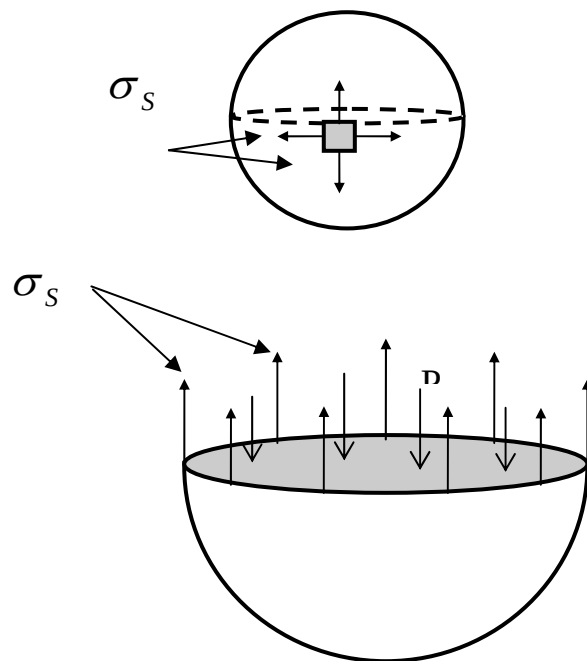
1. The variation in tangential stresses through the wall thickness is small.
2. The radial stress can be neglected. As the radial stress is small in comparison with the circumferential and longitudinal stresses.

Stresses in thin walled pressure vessels

Consider a thin walled spherical container. The state of stress in the wall of the sphere must be such that there is the same direct stress, σ_s in every direction from any point, due to symmetry of sphere!

To determine σ_s we need some way of applying the equations of static equilibrium.

Imagine one half of the sphere, together with the compressed gas that it contains, cut off from the rest by a diametral plane, as shown below:



Now consider the equilibrium of the composite “body” we have created. The pressure P required to confine the gas in this lower hemisphere acts in a downward direction, on a circular area πr^2 . Opposing this is a tensile stress σ_s , acting on the ring of radius R and width t_s , in which the diametral plane cuts the shell. To maintain vertical equilibrium, the downward force due to the pressure must equal the upward force due to the stress.

Therefore:

$$P \times (\pi R^2) = \sigma_s \times 2\pi R t_s$$

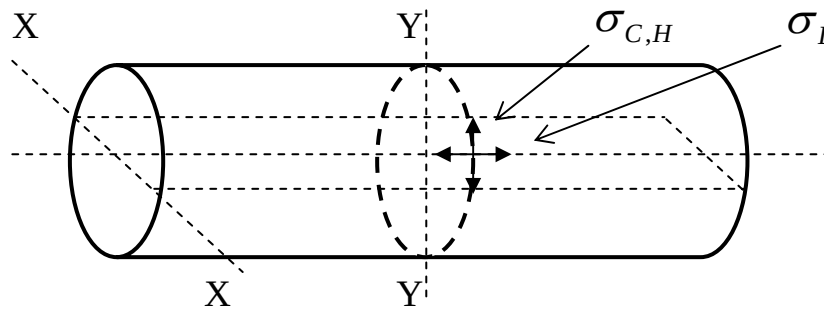
Giving:

$$\sigma_s = \frac{PR}{2t_s} \quad \text{or} \quad \frac{PD}{4t}$$

Consider a thin walled cylindrical container

When a thin cylinder is subjected to internal pressure, stresses are induced in the walls of the cylinder on:

1. the longitudinal section XX, due to the force tending to separate the top and bottom halves.
2. the circumferential section YY, due to the force tending to separate the right and left ends of the cylinder.



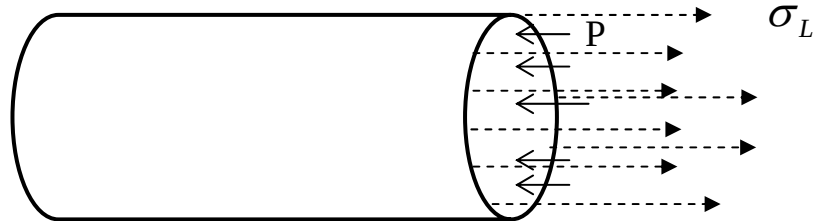
The stress in the longitudinal direction is called the Longitudinal Stress, and acts on the circumferential section

The Circumferential Stress acts on the longitudinal section, in the circumferential direction.

The Circumferential Stress is also known as the Hoop stress.

Firstly consider the stress in the longitudinal direction.

As we did with the sphere, make an imaginary section, as shown. The forces, which must balance to give equilibrium for the free body (horizontal), are:



- The force tending to separate the right and left-hand halves, is given by the pressure, P , multiplied by the area of one end:

$$P \times \frac{\pi d^2}{4}$$

Where d is the internal diameter.

- The force due to the longitudinal stress, σ_L , on the circumferential section is given by the stress multiplied by the cross sectional area of the wall (approx. $\pi \times d \times t$):

$$\sigma_L \times \pi dt$$

Therefore,

$$\sigma_L = \frac{P \times \frac{\pi d^2}{4}}{\pi dt}$$

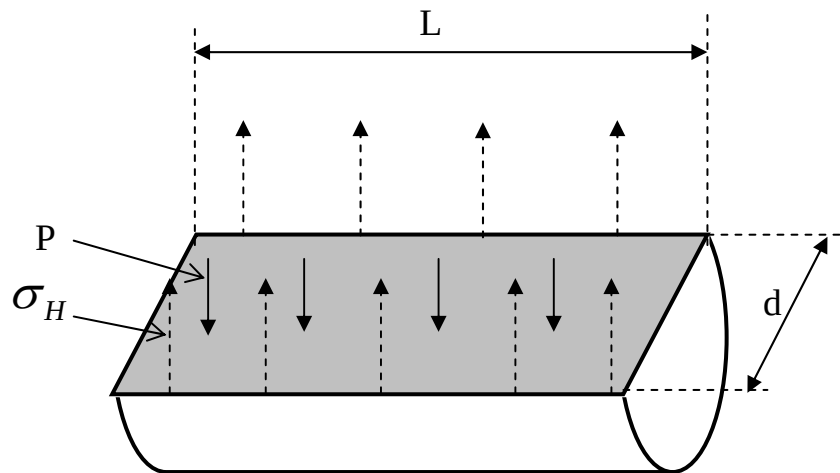
$$\therefore \sigma_L = \frac{Pd}{4t}$$

Now consider the stress in the circumferential direction.

Once again make an imaginary section, as shown. The force tending to separate the top and bottom halves is the pressure multiplied by the projected area.

In a direction perpendicular to the diametral plane this is given by:

$$P \times dL$$



This is resisted by the stress acting on the longitudinal section, σ_H . The force due to σ_H is:

$$\sigma_H \times 2tL$$

Therefore,

$$\sigma_H = \frac{PdL}{2tL}$$

$$\therefore \sigma_H = \frac{Pd}{2t}$$

Example 1

A closed cylindrical pressure vessel, with plane ends, is made from 3mm thick steel plate. The cylinder is 0.6m long and has an internal diameter of 0.2m. For an internal pressure of 2.5MN/m^2 , calculate the longitudinal and hoop stresses in the wall of the pressurised vessel.

Take Young's Modulus, $E = 210\text{GN/m}^2$ and Poisson's Ratio, $\nu = 0.286$

Thickness = $3 \times 10^{-3} \text{ m}$

Length = 0.6 m

$d = 0.2 \text{ m}$

Pressure, $P = 2.5 \times 10^6 \text{ N/m}^2$

$$\sigma_L = \frac{PD}{4t} = \frac{2.5 \times 10^6 \times 0.2}{4 \times 3 \times 10^{-3}} = 41.67 \text{ MN/m}^2$$

$$\sigma_H = \frac{PD}{2t} = 83.33 \text{ MN/m}^2$$

Determination of the associated strains

Consider strain in a spherical vessel:

The strain in any one circumferential direction through a point on the mid-surface of the sphere has two contributions:

- A contribution due to the direct tensile stress applied in the direction in question. If the sphere is of 'Hookean' material with Young's Modulus, E the first contribution is $\frac{\sigma_s}{E}$, an extension.
- A contribution due to the Poisson's Ratio effect.

There is an applied tensile stress σ_s in the direction at right angles to the one we are considering, this stress produces an extensional strain $\frac{\sigma_s}{E}$ in its direction of action resulting in a strain of $-\frac{\nu\sigma_s}{E}$ (a contraction) in the transverse direction - which is the one we are concerned with. The two contributions considered together give the circumferential strain:

$$\varepsilon_s = \frac{\sigma_s}{E} - \frac{\nu\sigma_s}{E} = \frac{\sigma_s}{E}(1 - \nu)$$

and substituting in $\sigma_s = \frac{PR}{2t_s}$ we can write,

$$\varepsilon_s = \frac{PR}{2t_s E}(1 - \nu)$$

Consider the strains in a cylinder

Using the same approach used above for the sphere:

$$\varepsilon_L = \frac{\sigma_L}{E} - \frac{\nu \sigma_H}{E}$$

but: $\sigma_H = \frac{PR}{t}$ and $\sigma_L = \frac{PR}{2t}$

We can therefore write:

$$\varepsilon_L = \frac{PR}{2tE} - \frac{\nu PR}{tE} = \frac{PR}{2tE} (1 - 2\nu)$$

The hoop, or circumferential, strain is similarly given by:

$$\varepsilon_H = \frac{\sigma_H}{E} - \frac{\nu \sigma_L}{E} = \frac{PR}{tE} - \frac{\nu PR}{2tE}$$

That is:

$$\varepsilon_H = \frac{PR}{2tE} (2 - \nu)$$

Compatibility of strains

If we now consider a cylindrical pressure vessel, but with hemi-spherical ends, a designer would have to be aware of the following point.

For the sphere, or a hemi-sphere,

$$\varepsilon_s = \frac{PR}{2t_s E}(1-\nu) \quad t_s = \text{thickness of sphere}$$

For the cylinder,

$$\varepsilon_H = \frac{PR}{2t_C E}(2-\nu) \quad t_C = \text{thickness of cylinder}$$

Unless these strains are equal, as the pressure builds up in the composite vessel, forces and stresses will arise not taken care of in our calculations. This problem can be eliminated at the design stage by ensuring: $\varepsilon_s = \varepsilon_H$

The condition can therefore be written as:

$$\therefore \frac{t_C}{t_s} = \left(\frac{2-\nu}{1-\nu} \right)$$

Change in Volume of a Cylinder

In order to determine the change in volume of a cylinder consider the following:

Circumference of cylinder changes by small amount $2\pi R\varepsilon_H$

Increase in mean radius, is therefore $R\varepsilon_H$

Increase in length of unit length of cylinder is ε_L

So change in the internal volume of a unit length of cylinder is;

$$\delta V = \pi(R + R\varepsilon_H)^2(1 + \varepsilon_L) - \pi R^2$$

$$\therefore \varepsilon_V = \frac{\delta V}{V} = (1 + \varepsilon_H)^2(1 + \varepsilon_L) - 1$$

But ε_H and ε_L are small quantities so,

$$(1 + \varepsilon_H)^2(1 + \varepsilon_L) - 1 = (2\varepsilon_H + \varepsilon_L)$$

Example 2

Considering again the problem in Example 1, calculate:

- the increase in length
- the increase in diameter
- the increase in volume of the vessel
- the decrease in thickness of the shell material.

Ignore any stiffening effects due to the ends.

Firstly, to find the increase in length:

$$\varepsilon_L = \frac{\sigma_L}{E} - \frac{\nu\sigma_H}{E} = 8.49 \times 10^{-5}$$

$$\varepsilon_L = \frac{\Delta L}{L} \quad \therefore \Delta L = 5.094 \times 10^{-5} m$$

Then, to find the increase in diameter:

$$\varepsilon_H = \frac{\sigma_H}{E} - \frac{\nu\sigma_L}{E} = 3.40 \times 10^{-4}$$

$$\varepsilon_H = \frac{\Delta D}{D} \quad \therefore \Delta D = 6.8 \times 10^{-5} m$$

Then, to find the increase in volume:

$$\varepsilon_v = 2.\varepsilon_H + \varepsilon_L = 7.64 \times 10^{-4}$$

So increase in volume = $V \times \varepsilon_v = 1.44 \times 10^{-5} m^3$

And change in thickness from:

$$\varepsilon_R = - \frac{\nu\sigma_H}{E} - \frac{\nu\sigma_L}{E} = -1.702 \times 10^{-4}$$

$$\Delta t = 3 \times 10^{-3} \times (-1.702 \times 10^{-4}) = -5.107 \times 10^{-4} m$$

This means a decrease of $5.1 \times 10^{-4} mm$

Tutorials

Question 1

A requirement normally written into the specification for a commercial aircraft is:

“The internal cabin pressure shall be maintained at a minimum of 70kN/m^2 (corresponding to the air pressure at 10,000 feet) at the maximum operating altitude”

For a particular commercial aircraft, just capable of reaching 50,000 feet (ambient air pressure = 12kN/m^2) the maximum differential is around 60kN/m^2 .

The fuselage of a commercial aircraft may be modelled as a **thin** cylindrical pressure vessel. During a routine flight the fuselage is subjected to an internal (differential) pressure of 60kN/m^2 . The mean diameter of the fuselage may be taken as 5m, with a corresponding shell thickness of 5mm.

Take Poisson’s Ratio = 0.3 and Young’s Modulus, $E = 70 \times 10^9 \text{N/m}^2$

Determine the change in diameter of the fuselage.

When you are happy that this is correct consider the following:

Determine the change in volume of the fuselage.

Determine the overall change in length.

If the cylindrical model is now refined slightly to include two hemispherical domed ends, determine the thickness of dome required in order to ensure compatibility of strains at the cylinder/dome interface.

Question 2

A thin steel cylinder of internal diameter 200mm, is subjected to a constant internal pressure of 5 MN/m^2 . The maximum allowable stress in the cylinder wall is 120 MN/m^2 .

Find:

- the required wall thickness
- change in diameter
- change in wall thickness

An axial load of 200kN is then applied to the cylinder, in addition to the internal pressure of 5 MN/m^2 . Find the new wall thickness required to maintain the maximum allowable stress limit.

State any assumptions made in your calculations.

Take Poisson's Ratio = 0.3 and Young's Modulus, $E = 207 \times 10^9 \text{ N/m}^2$

Question 3

A thin cylindrical shell has an internal diameter of 20 cm, and is 0.5 cm thick. It is subjected to an internal pressure of 3.5 MN/m^2 . Calculate the circumferential and longitudinal stresses if the ends of the cylinder are closed.

(Ans 35 & 70 MN/m^2)

Question 4

Air at 40 bar (pressure) is to be contained in the cylinder of a brake mechanism. If the stress is not to exceed 120 MN/m^2 and the internal diameter is 140 mm, find the required thickness?

(Ans 11.7 & 23.3mm)

Question 5

An air vessel, which is made of steel, is 2m long. It has a mean diameter of 45cm and is 1cm thick. Find the increase of mean diameter and the increase of length when charged to an internal pressure of 10 MN/m^2 .

For steel: $E = 200 \text{ GN/m}^2$, $\nu = 0.3$ (Ans $4.32 \times 10^{-4} \text{ m}$)